

Statistics & Probability

ANSWER KEY



Mean, median, and mode

- Find the mean of 4, 8, 6, 10, 7.
Sum = 35. Mean = $\frac{35}{5} = 7$.
 - Find the median of 3, 9, 5, 1, 7.
Sorted: 1, 3, 5, 7, 9. Median = 5.
 - Find the mode of 2, 4, 4, 6, 8, 4, 2.
4 appears three times, more than any other value, so the mode is 4.
 - A data set has values 8, 12, 10, 15, 10, 14, 11. Find the mean, median, and mode.
Sum = 80, mean = $\frac{80}{7} \approx 11.43$. Sorted: 8, 10, 10, 11, 12, 14, 15; median (4th value) = 11. Mode = 10 (appears twice).
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Range and quartiles

- Find the range of 5, 12, 8, 20, 3.
Range = $20 - 3 = 17$.
 - For the data set 2, 4, 5, 7, 8, 9, 12, find Q_1 and Q_3 .
Median (4th value) = 7. Lower half 2, 4, 5: $Q_1 = 4$. Upper half 8, 9, 12: $Q_3 = 9$.
 - Find the interquartile range (IQR) for the same data set.
 $IQR = Q_3 - Q_1 = 9 - 4 = 5$.
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Variance and standard deviation

- Find the variance of 2, 4, 6, 8, 10, using $\sigma^2 = \frac{\sum (x - \mu)^2}{n}$.
Mean = 6. Squared deviations: 16, 4, 0, 4, 16, sum = 40. Variance = $\frac{40}{5} = 8$.
 - Find the standard deviation of the same data set, rounded to two decimal places.
 $\sigma = \sqrt{8} \approx 2.83$.
 - A data set has variance 25. State its standard deviation.
 $\sigma = \sqrt{25} = 5$.
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Frequency tables

- 11** A frequency table shows value 1 with frequency 3, value 2 with frequency 5, and value 3 with frequency 2. Find the mean.

$$\text{Sum} = 1(3) + 2(5) + 3(2) = 3 + 10 + 6 = 19. \text{ Total frequency} = 10. \text{ Mean} = \frac{19}{10} = 1.9.$$

- 12** Using the same frequency table, find the mode.

Value 2 has the highest frequency (5), so the mode is 2.

- 13** Using the same frequency table, find the median.

There are 10 values in order 1, 1, 1, 2, 2, 2, 2, 2, 3, 3. Median = $\frac{2 + 2}{2} = 2$ (average of the 5th and 6th values).

Basic probability

- 14** A fair six-sided die is rolled. Find $P(\text{number} > 4)$.

$$\text{Favorable outcomes } \{5, 6\}: P = \frac{2}{6} = \frac{1}{3}.$$

- 15** A bag contains 5 red and 3 blue marbles. Find $P(\text{blue})$.

$$P(\text{blue}) = \frac{3}{8}.$$

- 16** A card is drawn from a standard 52-card deck. Find $P(\text{king})$.

$$P(\text{king}) = \frac{4}{52} = \frac{1}{13}.$$

- 17** Two fair coins are flipped. Find $P(\text{both heads})$.

$$P = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

Complementary and mutually exclusive events

- 18** If $P(A) = 0.35$, find $P(\text{not } A)$.

$$P(\text{not } A) = 1 - 0.35 = 0.65.$$

- 19** A die is rolled. Events $A = \{\text{even}\}$ and $B = \{\text{rolling a } 5\}$. Are A and B mutually exclusive?

Yes — $\{2, 4, 6\}$ and $\{5\}$ share no outcomes.

- 20** If $P(A) = 0.4$ and $P(B) = 0.25$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

$$P(A \text{ or } B) = 0.4 + 0.25 = 0.65.$$

Independent events

- 21** Two independent events have $P(A) = 0.6$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

$$P(A \text{ and } B) = 0.6 \times 0.5 = 0.3.$$

- 22 A fair coin is flipped twice. Find $P(\text{first heads and second tails})$.
 $P = 0.5 \times 0.5 = 0.25$.
- 23 A card is drawn from a deck, replaced, then a second card is drawn. Find $P(\text{both are aces})$.
 $P = \frac{4}{52} \times \frac{4}{52} = \frac{1}{13} \times \frac{1}{13} = \frac{1}{169}$.
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Conditional probability

- 24 In a class, $P(\text{math}) = 0.7$ and $P(\text{physics and math}) = 0.4$. Find $P(\text{physics} \mid \text{math})$.
 $P(\text{physics} \mid \text{math}) = \frac{0.4}{0.7} \approx 0.571$.
- 25 A box has 10 balls: 6 red and 4 blue. Two balls are drawn without replacement. Find $P(\text{2nd is blue} \mid \text{1st is blue})$.
After removing one blue ball, 3 of the remaining 9 balls are blue: $P = \frac{3}{9} = \frac{1}{3}$.
- 26 Given $P(A) = 0.5$, $P(B) = 0.3$, $P(A \text{ and } B) = 0.15$, find $P(A \mid B)$.
 $P(A \mid B) = \frac{0.15}{0.3} = 0.5$.
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Counting: permutations and combinations

- 27 In how many ways can 4 books be arranged on a shelf?
 $4! = 24$ ways.
- 28 How many ways can a committee of 3 be chosen from 8 people (order doesn't matter)?
 $\binom{8}{3} = \frac{8!}{3!5!} = 56$.
- 29 A code uses 3 distinct digits from 0–9, where order matters. How many possible codes are there?
 $10 \times 9 \times 8 = 720$.
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The five-number summary

- 30 For the data set 6, 10, 14, 18, 22, 26, 30, find the five-number summary (minimum, Q_1 , median, Q_3 , maximum).
Median (4th value) = 18. Lower half 6, 10, 14: $Q_1 = 10$. Upper half 22, 26, 30: $Q_3 = 26$. Summary: 6, 10, 18, 26, 30.
- 31 Find the IQR for the same data set.
 $IQR = Q_3 - Q_1 = 26 - 10 = 16$.
- 32 Outlier boundaries are $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$. Find both boundaries for the same data set.
Lower: $10 - 1.5(16) = 10 - 24 = -14$. Upper: $26 + 1.5(16) = 26 + 24 = 50$.

Reading data from a bar chart

- 33** The bar chart shows the number of books sold each day at a bookstore over five days.
- a** Find the total number of books sold over the five days. $\text{Total} = 12 + 18 + 9 + 15 + 21 = 75$ books.
 - b** Find the mean number of books sold per day. $\text{Mean} = \frac{75}{5} = 15$ books per day.
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Word problems: statistics and probability in context

- 34** This question has two parts. A survey of 200 students found that 120 like math and 90 like science, while 50 like both subjects.
- a** Find the probability that a randomly selected student likes math or science, using $P(M \cup S) = P(M) + P(S) - P(M \cap S)$. $P(M) = \frac{120}{200} = 0.6$, $P(S) = \frac{90}{200} = 0.45$, $P(M \cap S) = \frac{50}{200} = 0.25$.
 $P(M \cup S) = 0.6 + 0.45 - 0.25 = 0.8$.
 - b** Find the probability that a student likes neither subject. $P(\text{neither}) = 1 - 0.8 = 0.2$.
- 35** This question has two parts. A quality-control inspector finds that 2% of items from a production line are defective. A random sample of 3 items is tested, and defects occur independently.
- a** Find the probability that all three items are defective. $P(\text{all defective}) = 0.02^3 = 0.000008$.
 - b** Find the probability that none of the three items are defective.
 $P(\text{none defective}) = 0.98^3 \approx 0.9412$.
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MCQ — Mean, median, and mode

- 36** Find the mean of 5, 9, 7, 11, 8.

a. 8 ✓

b. 7

c. 40

d. 9

Sum = 40; mean = $40 \div 5 = 8$.

37 Find the median of 4, 10, 6, 2, 8.

a. 5

b. 8

c. 6 ✓

d. 4

Sorted: 2, 4, 6, 8, 10; median = 6.

38 Find the mode of 3, 5, 5, 7, 9, 5, 3.

a. 5 ✓

b. 4

c. 3

d. 7

5 appears 3 times, more than any other value.

39 A data set has values 9, 13, 11, 16, 11, 15, 12. Find the mean, median, and mode.

a. Mean = 13, median = 12, mode = 11

b. Mean = 12.43, median = 11, mode = 12

c. Mean $\approx$ 12.43, median = 12, mode = 11 ✓

d. Mean $\approx$ 12.43, median = 13, mode = 11

Sum = 87, mean = $87 \div 7 \approx 12.43$. Sorted: 9, 11, 11, 12, 13, 15, 16; median = 12; mode = 11.

40 Find the mean of 12, 18, 15, 21, 14.

a. 15

b. 16 ✓

c. 80

d. 18

Sum = 80; mean = $80 \div 5 = 16$.

MCQ — Range and quartiles

41 Find the range of 6, 14, 9, 22, 4.

- a. 22
- b. 16
- c. 18 ✓
- d. 20

$$22 - 4 = 18.$$

42 For the data set 3, 5, 6, 8, 9, 10, 13, find Q_1 and Q_3 .

- a. $Q_1 = 5, Q_3 = 9$
- b. $Q_1 = 5, Q_3 = 10$ ✓
- c. $Q_1 = 3, Q_3 = 13$
- d. $Q_1 = 6, Q_3 = 9$

Lower half 3, 5, 6: $Q_1 = 5$; upper half 9, 10, 13: $Q_3 = 10$.

43 Find the interquartile range (IQR) for the same data set.

- a. 10
- b. 8
- c. 3
- d. 5 ✓

$$IQR = Q_3 - Q_1 = 10 - 5 = 5.$$

44 Find the range of 18, 25, 12, 30, 9.

- a. 18
- b. 16
- c. 21 ✓
- d. 30

$$30 - 9 = 21.$$

MCQ — Variance and standard deviation

45 Find the variance of 3, 5, 7, 9, 11, using $\sigma^2 = \frac{\sum (x - \mu)^2}{n}$.

- a. 4
- b. 2.83
- c. 8 ✓
- d. 40

Mean = 7; deviations squared sum to 40; variance = $40 \div 5 = 8$.

46 Find the standard deviation of the same data set, rounded to two decimal places.

- a. 1.41
- b. 4.00
- c. 2.83 ✓
- d. 8.00

$$\sqrt{8} \approx 2.83.$$

47 A data set has variance 36. State its standard deviation.

- a. 6 ✓
- b. 12
- c. 18
- d. 1,296

$$\sqrt{36} = 6.$$

48 Find the variance of 4, 8, 12, 16, 20, using $\sigma^2 = \frac{\sum (x - \mu)^2}{n}$.

- a. 160
- b. 5.66
- c. 16
- d. 32 ✓

Mean = 12; deviations squared sum to 160; variance = $160 \div 5 = 32$.

MCQ — Frequency tables

49 A frequency table shows value 1 with frequency 4, value 2 with frequency 6, and value 3 with frequency 5. Find the total number of data points.

- a. 10
- b. 15 ✓
- c. 3
- d. 6

$$4 + 6 + 5 = 15.$$

50 Using the same frequency table, find the mode.

a. 6

b. 3

c. 2 ✓

d. 1

Value 2 has the highest frequency (6).

51 Using the same frequency table, find the median.

a. 1.5

b. 1

c. 2 ✓

d. 3

With 15 values, the median is the 8th, which falls in the value-2 group.

52 Using the same frequency table, find the mean.

a. 2.33

b. ≈ 2.07 ✓

c. 2.00

d. 1.87

Sum = $1(4) + 2(6) + 3(5) = 31$; mean = $31 \div 15 \approx 2.07$.

53 The bar chart shows a frequency table. Find the total number of data points.

a. 15 ✓

b. 6

c. 3

d. 10

$4 + 6 + 5 = 15$.

MCQ — Basic probability

54 A fair six-sided die is rolled. Find $P(\text{number} < 3)$.

a. $\frac{1}{6}$

b. $\frac{1}{2}$

c. $\frac{1}{3}$ ✓

d. $\frac{2}{3}$

$\{1, 2\}: \frac{2}{6} = \frac{1}{3}.$

55 A bag contains 7 red and 4 blue marbles. Find $P(\text{red})$.

a. $\frac{7}{11}$ ✓

b. $\frac{1}{11}$

c. $\frac{4}{11}$

d. $\frac{7}{4}$

7 out of 11 total.

56 A card is drawn from a standard 52-card deck. Find $P(\text{queen})$.

a. $\frac{4}{13}$

b. $\frac{1}{4}$

c. $\frac{1}{13}$ ✓

d. $\frac{1}{52}$

4 queens out of 52: $\frac{1}{13}.$

57 Two fair coins are flipped. Find $P(\text{at least one tail})$.

a. $\frac{1}{2}$

b. $\frac{3}{4}$ ✓

c. $\frac{1}{4}$

d. $\frac{1}{3}$

$P(\text{no tails}) = \frac{1}{4}$, so $P(\text{at least one}) = \frac{3}{4}$.

58 A fair six-sided die is rolled. Find $P(\text{multiple of 3})$.

a. $\frac{2}{5}$

b. $\frac{1}{3}$ ✓

c. $\frac{1}{2}$

d. $\frac{1}{6}$

$\{3, 6\}: \frac{2}{6} = \frac{1}{3}$.

MCQ — Complementary and mutually exclusive events

59 If $P(A) = 0.28$, find $P(\text{not } A)$.

a. 0.28

b. 1.28

c. 0.5

d. 0.72 ✓

$1 - 0.28 = 0.72$.

60 A die is rolled. Events $A = \{\text{odd}\}$ and $B = \{\text{rolling a 4}\}$. Are A and B mutually exclusive?

a. Yes ✓

b. Only sometimes

c. No

d. Cannot be determined

$A = \{1, 3, 5\}$ and $B = \{4\}$ share no outcomes: yes.

61 If $P(A) = 0.3$ and $P(B) = 0.45$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

a. 0.75 ✓

b. 0.15

c. 1.05

d. 0.135

$0.3 + 0.45 = 0.75$.

62 If $P(A) = 0.6$, find $P(\text{not } A)$.

a. 1.6

b. 0.4 ✓

c. 0.5

d. 0.6

$1 - 0.6 = 0.4$.

MCQ — Independent events

63 Two independent events have $P(A) = 0.7$ and $P(B) = 0.4$. Find $P(A \text{ and } B)$.

a. 1.1

b. 0.28 ✓

c. 0.3

d. 0.175

$$0.7 \times 0.4 = 0.28.$$

64 A fair coin is flipped twice. Find $P(\text{both heads})$.

a. $\frac{1}{3}$

b. 1

c. $\frac{1}{4}$ ✓

d. $\frac{1}{2}$

$$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

65 A card is drawn from a deck, replaced, then a second card is drawn. Find $P(\text{both are kings})$.

a. $\frac{2}{13}$

b. $\frac{1}{26}$

c. $\frac{1}{13}$

d. $\frac{1}{169}$ ✓

$$\left(\frac{4}{52}\right)^2 = \frac{1}{169}.$$

66 Two independent events have $P(A) = 0.5$ and $P(B) = 0.8$. Find $P(A \text{ and } B)$.

a. 0.65

b. 1.3

c. 0.3

d. 0.4 ✓

$$0.5 \times 0.8 = 0.4.$$

MCQ — Conditional probability

67 In a class, $P(\text{science}) = 0.65$ and $P(\text{math and science}) = 0.3$. Find $P(\text{math} \mid \text{science})$.

a. 0.65

b. 0.195

c. ≈ 0.46 ✓

d. 0.3

$$\frac{0.3}{0.65} \approx 0.46.$$

68 A box has 12 balls: 7 red and 5 blue. Two balls are drawn without replacement. Given the first is blue, find $P(\text{second is blue})$.

a. $\frac{5}{11}$

b. $\frac{4}{11}$ ✓

c. $\frac{5}{12}$

d. $\frac{4}{12}$

After removing one blue, 4 blue remain out of 11 total.

69 Given $P(A) = 0.6$, $P(B) = 0.4$, $P(A \text{ and } B) = 0.2$, find $P(A \mid B)$.

a. 0.3

b. 0.33

c. 0.8

d. 0.5 ✓

$$\frac{0.2}{0.4} = 0.5.$$

70 Given $P(A) = 0.45$, $P(B) = 0.5$, $P(A \text{ and } B) = 0.2$, find $P(B | A)$.

a. 0.4

b. 0.09

c. 0.5

d. ≈ 0.44 ✓

$$\frac{0.2}{0.45} \approx 0.44.$$

MCQ — Counting: permutations and combinations

71 In how many ways can 5 books be arranged on a shelf?

a. 25

b. 120 ✓

c. 20

d. 60

$$5! = 120.$$

72 How many ways can a committee of 4 be chosen from 9 people (order doesn't matter)?

a. 126 ✓

b. 3,024

c. 362,880

d. 36

$$\binom{9}{4} = 126.$$

73 A code uses 4 distinct digits from 0–9, where order matters. How many possible codes are there?

- a. 10,000
- b. 40
- c. 210
- d. 5,040 ✓

$$10 \times 9 \times 8 \times 7 = 5,040.$$

74 In how many ways can 6 people be arranged in a line?

- a. 720 ✓
- b. 120
- c. 36
- d. 6

$$6! = 720.$$

MCQ — The five-number summary

75 For the data set 5, 9, 13, 17, 21, 25, 29, find the five-number summary (minimum, Q_1 , median, Q_3 , maximum).

- a. 9, 13, 17, 21, 25
- b. 5, 9, 17, 25, 29 ✓
- c. 5, 13, 17, 21, 29
- d. 5, 9, 13, 25, 29

$$\text{Min} = 5; Q_1 = 9; \text{median} = 17; Q_3 = 25; \text{max} = 29.$$

76 Find the IQR for the same data set.

- a. 8
- b. 24
- c. 16 ✓
- d. 20

$$IQR = Q_3 - Q_1 = 25 - 9 = 16.$$

77 Outlier boundaries are $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$. Find both boundaries for the same data set.

- a. -15 and 49 ✓
- b. 9 and 25
- c. -15 and 40
- d. -24 and 40

$$9 - 1.5(16) = -15; 25 + 1.5(16) = 49.$$

78 For the data set 10, 15, 20, 25, 30, 35, 40, find the median.

- a. 25 ✓
- b. 30
- c. 20
- d. 27.5

With 7 values sorted, the median is the 4th: 25.

MCQ — Reading data from a bar chart

79 The bar chart shows the number of books sold each day at a bookstore over five days. Find the total books sold.

a. 87 ✓

b. 22

c. 77

d. 97

$$12 + 18 + 15 + 22 + 20 = 87.$$

80 Using the same bar chart, find the mean number of books sold per day.

a. 15

b. 18

c. 20

d. 17.4 ✓

$$87 \div 5 = 17.4.$$

81 Using the same bar chart, find the range of books sold.

a. 12

b. 22

c. 8

d. 10 ✓

$$22 - 12 = 10.$$

82 Using the same bar chart, on which day were the most books sold?

a. Wednesday

b. Thursday ✓

c. Tuesday

d. Friday

Thursday shows the tallest bar, 22 books.

MCQ — Word problems: statistics and probability in context

- 83** A survey of 250 students found that 150 like math and 100 like science, with 40 liking both. Find the probability that a randomly selected student likes math or science.

a. 0.84 ✓

b. 0.6

c. 0.16

d. 1.0

$$\frac{150 + 100 - 40}{250} = \frac{210}{250} = 0.84.$$

- 84** A quality-control inspector finds that 3% of items from a production line are defective. If 500 items are inspected, how many are expected to be defective?

a. 150

b. 15 ✓

c. 3

d. 30

$$0.03 \times 500 = 15.$$

- 85** In a survey of 180 people, 108 prefer coffee over tea. Find the probability, as a percent, that a randomly selected person prefers coffee.

a. 60% ✓

b. 50%

c. 108%

d. 40%

$$\frac{108}{180} = 60\%.$$